🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.4

Represent Three-Dimensional Figures in Two Dimensions

Lesson 5-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the surface area of a solid by using its net.
Language Objective: I can explain my work using the words surface area, net, face, and two-dimensional.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Surface Area Using Nets | Surface area | Net | Face | Two-dimensional | Edge

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. The total area of every face, found by adding the faces in a net, is the ________.
2. The total area of all the faces of a solid is its ________.
3. A flat pattern that folds up into a solid is a ________.
4. A flat surface of a solid figure is a ________.
5. A flat shape with length and width but no thickness is ________.
6. A line segment where two faces meet is an ________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How many sheets of wrapping paper are needed to cover the entire box?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Surface Area Using Nets** — The total area of every face of a three-dimensional figure, found by adding the faces shown in its net. Formula: SA = the sum of the areas of all the faces.
  Área de superficie usando redes — El área total de todas las caras de una figura tridimensional, hallada al sumar las caras que muestra su plantilla. Fórmula: SA = la suma de las áreas de todas las caras.
- ☐ **Surface area** — The total area of all the flat sides of a solid.
  Área de superficie — El área total de todos los lados planos de un sólido.
- ☐ **Net** — A flat shape that folds up into a solid.
  Plantilla (desarrollo plano) — Una figura plana que se dobla y forma un sólido.
- ☐ **Face** — One flat side of a solid shape.
  Cara — Un lado plano de una figura sólida.
- ☐ **Two-dimensional** — A flat shape with length and width, but no thickness.
  Bidimensional — Una figura plana con largo y ancho, pero sin grosor.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The box's surface area is ____ in² because SA = 2(____×____) + 2(____×____) + 2(____×____) = ____ + ____ + ____ = ____. You need ____ sheets because ____ ÷ 400 ≈ ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A net lays every face flat so you can see and measure all six faces.

Evidence: Students explain a rectangular prism has 6 faces, and the net unfolds the solid into a flat (two-dimensional) pattern so every face is visible to measure.

Reasoning: This evidence connects the definition of surface area using nets to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The box's surface area is ____ in² because SA = 2(____×____) + 2(____×____) + 2(____×____) = ____ + ____ + ____ = ____. You need ____ sheets because ____ ÷ 400 ≈ ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Surface Area Using Nets
2. **Notes 2:** Surface area
3. **Notes 3:** Net
4. **Notes 4:** Face
5. **Notes 5:** Two-dimensional
6. **Notes 6:** Edge
7. **Watch for this mistake:** *A common mistake in Surface Area Using Nets is finding the area of only one face from each of the 3 matching pairs and forgetting to double it, instead of counting all 6 faces. For a box that is 9 in × 4 in × 5 in, a student might add just one face from each pair: (9×4) + (9×5) + (4×5) = 36 + 45 + 20 = 101 in², but the net actually has two of each face, so the correct total is 2(36) + 2(45) + 2(20) = 72 + 90 + 40 = 202 in² — exactly double. Before you submit, ask: 'Did I count both matching faces in each pair, or only one from each pair?'*
8. **Try It 1:** A. 52 in² *— From the net: SA = 2(2×3) + 2(2×4) + 2(3×4) = 12 + 16 + 24 = 52 in².*
9. **Try It 2:** *— When a net folds up, each flat shape becomes one face of the solid, so the number and the shapes of the faces identify the solid — six identical squares fold into a cube, while three pairs of matching rectangles fold into a rectangular prism.*